

THE OHIO Naturalist

PUBLISHED BY
THE BIOLOGICAL CLUB OF THE OHIO STATE UNIVERSITY

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Volume 1.

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Number 3

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PRESS OF HANN & ADAIR

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CONTENTS

Notes on the Self-pruning of Trees	29
<i>John H. Schaffner and Fred. J. Tyler</i>	
Plant Names Commemorative of Ohio Botanists	33
<i>Clara Armstrong</i>	
The Maximum Height of Plants II.	39
<i>John H. Schaffner</i>	
Campus Birds	40
<i>Robert F. Griggs</i>	
Minor Plant Notes No. 1	46

The Ohio Naturalist

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No. 3

NOTES ON THE SELF-PRUNING OF TREES.

JOHN H. SCHAFFNER AND FRED J. TYLER.

In a dense forest of growing trees the smaller, side branches of the main stem, as well as those of the larger branches, are continually dying off. But the tree rids itself of these dead branches by forming a collar of tissue from the cambium layer around the base of the branch, which presses more tightly as layer after layer of living wood is added, until the branch finally falls off and the hole which is left is grown over in a short time. This process is known as natural pruning. But the process which we wish to consider is very different from this, and we desire to distinguish it by the term, self-pruning. In this case the living branches are cut off or else the

cutting-off process is the cause of the death of the branch. A special adaptation is provided to accomplish the result and the process is one whose purpose is the shedding of the branches rather than the attempt to accommodate the plant to conditions of injury brought about by other causes. In a number of species perfectly formed winter buds were developed on the branches which were shed, and so far as our observations go, the twigs are cast in the fall and winter.

Although the shedding of branches is well known, especially in the conifers, not as much notice has been taken of it as we think it deserves. We have been taking observations for several years and have been partly anticipated by Dr. Bessey in a note in *Science* 12. 650, 1900,—Botanical Notes—The Annual Shedding of Cottonwood Twigs. Bessey describes the shedding of the twigs of *Populus deltoides* as occurring about the middle of October, and after giving the details of the process, concludes as follows: “It is an interesting fact that the Tamarisks (*Tamarix* sp.) which are held by some botanists to be closely related to the Poplars, shed their twigs by exactly the same device as that described above. In the Tamarisks the shedding of the twigs is a part of the annual process of defoliation, their leaves being so small that it appears to be less trouble and expense to drop twig and all than to separate every individual leaf. Possibly in the Cottonwoods, with their large leaves, we have a survival of the Tamarisk twig-shedding habit long after its original significance has disappeared.” Dr. Bessey, however, we believe, will not insist on this supposition when he considers that the same thing occurs in species of *Prunus*, *Quercus*, and other widely separated genera.

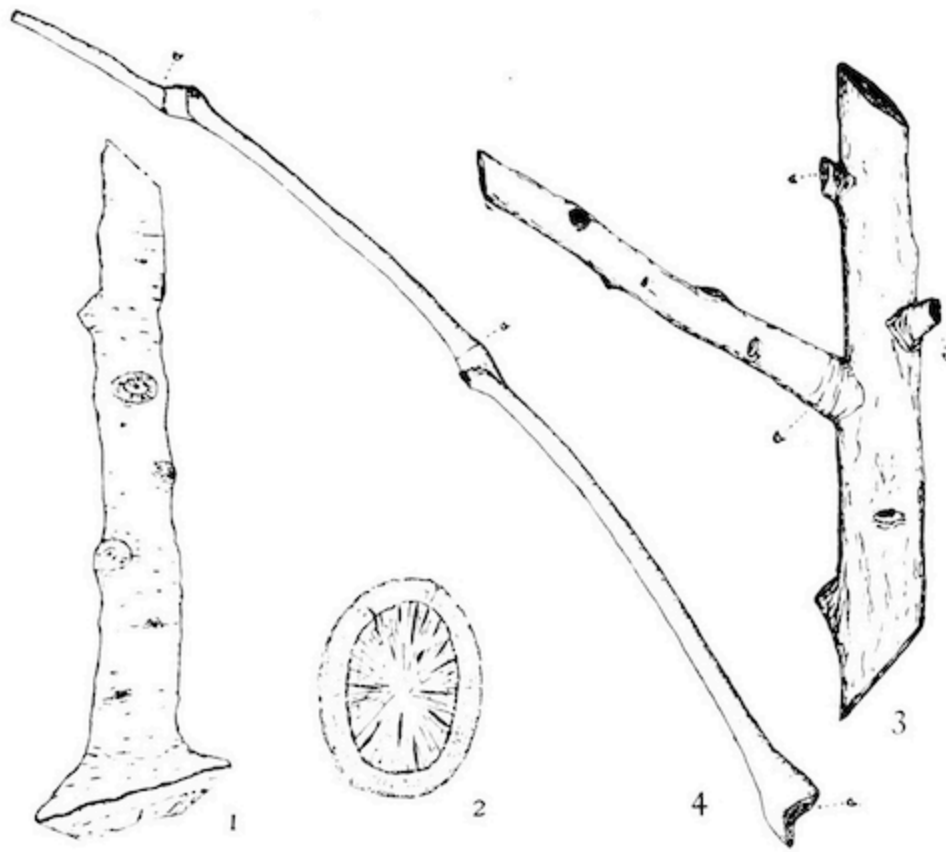


FIG. 1—Twig of *Populus alba*, showing large basal joint and scars where smaller twigs have been detached.

FIG. 2—View of basal joint of same twig as Fig. 1.

FIG. 3—Twig of *Salix nigra*, showing position of the brittle zone (a).

FIG. 4—Green twig of *Ampelopsis cordata*, showing joints (a) at the leaf nodes.

The self-pruning of twigs from woody stems, so far as our observations go at present, is accomplished in three general ways: 1st, by the formation of a single joint close to the parent branch; 2nd, by the formation of a brittle zone near the base of the limbs which are to be shed; 3rd, by a series of transverse joints corresponding to the leaf nodes.

Among the plants which come in the first class, the most striking perhaps, is *Populus alba*, in which very large branches are frequently cut off in such a perfect manner that one might think the pruning had been done

with a sharp knife. Green twigs from one to fifteen years old were found to be shed and no doubt still older ones are cut off. The twigs have well-developed winter buds and this is also the case in other species of *Populus* and in certain species of *Quercus* and *Prunus*. In *Quercus alba* shed twigs were found from one to four years old while in *Q. acuminata* there were some seven years of age. In *Prunus serotina* twigs from one to six years old were cut off. In *Ulmus Americana* not only are joints formed at the base of the twigs, but the twigs also break apart at the nodes, caused by annual growth.

The following species were found belonging to the first class:

Populus alba L.

“ *deltoides* Marsh.

“ *grandidentata* Mx.

“ *tremuloides* Mx.

“ *dilatata* Ait.

Quercus alba L.

“ *robur* L.

“ *macrocarpa* Mx.

“ *acuminata* (Mx.) Sarg.

Ulmus americana, L.

Prunus serotina, Ehrh.

Tamarix gallica L.

Tsuga canadensis (L.) Carr.

All the plants found with brittle zones belonged to the willows. It is interesting to note that the branches shed may be one to several years of age and that certain branches do not develop a brittle zone. *Salix nigra* and *S. amygdaloides* seem to show the character most perfectly and it is remarkable to see how readily the branches drop off.

The following species show the adaptation:

Salix nigra Marsh.

“ *amygdaloides* anders.

“ *fragilis* L.

“ *alba vitellina* (L.) Koch.

“ *babylonica* L.

Among those which come in the third class, the most remarkable plant observed was *Ampelopsis cordata*. About the time that the leaves are shed nearly all the slender green branches literally fall to pieces and drop off. Most of the fruit is on these branches and the berries are thus shed at the same time. In the winter the plant is remarkable for the few branches left and it looks like an artificially pruned vine.

The species observed belonging to this class are the following:

Ampelopsis cordata Mx.

“ *tricuspidata* Seib. & Zucc.

Parthenocissus quinquefolia (L.) Planch.

The shedding of the twigs of woody plants may in many cases be entirely an adaptation to get rid of the leaves as in the case of the dwarf branches of Pines and the young twigs of Tamarisks. But even in the Tamarisks it is doubtful whether the joints formed in the older branches can be claimed to have such a purpose, since in this case the leaves have all been shed with the annual twigs. The shedding of the old woody branches may have a different purpose. In regard to the trees mentioned above, we think that the process is one primarily to rid the tree of surplus branches. This would manifestly be an advantage and would give room and opportunity for the development of many young leafy shoots every year without accumulating too great a mass of useless members. This is certainly the case with the poplars and the willows. In no case do we think it admissible to say that the adaptation is primarily for the purpose of propagation, although this may be a very important incidental result in such plants as the willows when growing in wet places. In the case of *Ampelopsis cordata*, the only reasonable explanation seems to be a preparation for the winter condition, since the branches which are shed do not ripen and the plant has an admirable method for shedding its leaves. The slender branches would be in great danger of being killed by the cold of winter. The shedding of the young branches of *Taxodium distichum* (L.) Rich. is remarkably like that in Tamarix. The slender, dwarf branches clothed with the leaves drop off in the fall or the following spring. The habit must be quite ancient, as such branches of *Taxodium distichum miocenum* Hr. are very abundantly preserved as fossils. In *Taxodium mexicanum* Carr. the dwarf branches are not shed until the second year. *Glyptostrobus*

pendulus Endl. and *G. heterophyllus* Endl. also have deciduous dwarf branches. There are other conifers and no doubt many other angiospermous trees which possess these interesting adaptations and by careful observations, no doubt many interesting ecological facts will be brought to light.

PLANT NAMES COMMEMORATIVE OF OHIO BOTANISTS.

CLARA ARMSTRONG.

Though Ohio has had fewer botanists than many other States, some of these became known wherever botany was cultivated. Riddell was one of the pioneer botanists of the west and for six years he was a resident of this State. By the publication of his *Synopsis* he contributed largely to Ohio Botany. The names of Sullivant and Lesquereux shed still greater luster on our state. Most of the work of the former and all of that of the latter was done in Ohio. Several others whose names are given below have been industrious students of our Flora, and have spent many years if not all their lives within our territory. It is a long and interesting list of botanical names which commemorates their scientific labors. In the preparation of this paper I have been aided materially by Prof. W. A. Kellerman.

JOHN LEONARD RIDDELL.—Born in Leyden, Mass., Feb. 20th, 1807; died in New Orleans, La., Oct. 7, 1863. He graduated from Rensseler Institute, Troy, N. Y. He came to Ohio in 1830 and became professor of Botany and adjunct Professor of Chemistry in the Medical College of Cincinnati. He was an enthusiastic and industrious botanist, and collected extensively in many parts of our State. Scarcely any of his specimens seem now to be in existence though he prepared sets for sale and accumulated a large herbarium. His most important publication was the *Synopsis of the Flora of the Western States*. He also published a *Supplementary Catalogue of Ohio Plants*. In 1836 he left Ohio, carrying his botanical specimens to New Orleans where he became professor in a Medical College; he was also in the employ of the government until his death. He furnished many notes and longer articles to scientific journals and was the author of many new species of plants. The following have been named in his honor:

Riddellia, synonym of Psilostrophe, a genus of the Compositae family.

Solidago Riddellii, a species of Golden Rod.

Senecio Riddellii, synonym of S. Douglasii, a species of Compositae.

WILLIAM STARLING SULLIVANT.—Born in Franklinton, Ohio, Jan. 18, 1803, died in Columbus, Ohio, April 30, 1873. He was educated at Ohio University and Yale. The death of his father at the time of his graduation prevented him from studying for one of the learned professions and he became a surveyor and practical engineer, which occupation he followed until late in life. During this time he collected and studied the plants of central Ohio, and in 1840 he began to pay particular attention to Mosses; this became the subject of his special study, and he was soon recognized as the most eminent bryologist that this country has ever produced.

In 1864 the degree of LL.D. was conferred on Sullivant by Gambier College. He was elected to membership in many scientific societies both in the United States and Europe. At his death his bryological books, collections and preparations of Mosses were given to the Gray Herbarium of Harvard University. The remainder of his botanical library was bequeathed to the Ohio State University then called the Agricultural and Mechanical College. His microscopes were given to Starling Medical College. Of this institution he was senior trustee. He was the author of many new species and the plants named for him by other botanists are as follows:

Sullivantia, a genus of the Saxifragas. The name first proposed by Gray (1840) for the species discovered by Sullivant was Saxifraga Sullivantii; two years later Gray established the genus Sullivantia and published the name of the plant in question as Sullivantia Ohionis. According to the rules of priority that name now stands as Sullivantia Sullivantii.

Discina Sullivantii, a species of discomycetous fungi.

Panus Sullivantii, a species of Agaricineae.

Lentinus Sullivantii, a species of Agaricineae.

Marasmius Sullivantii, a species of Agaricineae.

Psilocybe Sullivantii, a species of Agaricineae.

Boletus Sullivantii, a species of the Polyporeae.
Polystictus Sullivantii, a species of the Polyporeae.
Thelephora Sullivantii, a species of the Thelephoreae.
Cephalozia Sullivantii, a species of Liverwort.
Jubula Sullivantii, a species of Liverwort.
Kantia Sullivantii, a species of Liverwort.
Porella Sullivantii, a species of Liverwort.
Hypnum Sullivantii, a species of Moss.
Astomum Sullivantii, a species of Moss.
Amphoridium Sullivantii, a species of Moss.
Brachythecium Sullivantii, a species of Moss.
Bruchia Sullivantii, a species of Moss.
Cylindrothecium Sullivantii, a species of Moss.
Fontinalis Sullivantii, a species of Moss.
Macromitrium Sullivantii, a species of Moss.
Neckera Sullivantii, a species of Moss.
Phascum Sullivantii, a species of Moss.
Pleuridium Sullivantii, a species of Moss.
Schlotheimia Sullivantii, a species of Moss.
Eurhynchium Sullivantii, a species of Moss.
Hookeria Sullivantii, a species of Moss.
Zygodon Sullivantii, a species of Moss.
Sphagnum Sullivantianum, a species of Moss.
Asclepias Sullivantii, a species of Milkweed.
Lonicera Sullivantii, a species of Honeysuckle.
Carex Sullivantii, a species of Sedge.
Callipteridium Sullivantii, a fossil plant.

There were also two species of Mosses named for Mrs. Sullivant as follows:

Hypnum Sullivantiae, a pleurocarpous Moss.
Plagiothecium Sullivantiae, a pleurocarpous Moss.

LEO LESQUEREUX.—Born in Fleurier, Switzerland, Nov. 18, 1806, died in Columbus, Ohio, Oct. 25, 1889. On entering the Academy of Neuchatel he met Arnold Guyot and together they became devoted to natural science. In

1822 he went to Eisenach preparatory to entering the University of Berlin, supporting himself by teaching French. He was principal of a college at Chaux de Fonds but had to give this up on account of deafness. From this time he did engraving, made watch springs, and studied Mosses and fossil plants.

In 1848 Lesquereux came to America settling at Cambridge, where he assisted Louis Agassiz; but he soon removed to Columbus, Ohio, where he lived until his death. He was a worthy associate of William S. Sullivan to whom in fact he was indebted by the most commendable generosity. He was a paleo-botanist and a student of Mosses. Many plants are named for him including a large number of fossil forms.

Lesquerella, a genus of Cruciferae.

Lesquereuxia, synonym of Siphonostegia, a genus of Scrophulariaceae.

Hysterographium Lesquereuxii, a species of Pyrenomycetous Fungi.

Webera Lescuriana, a species of Moss.

Bryum Lescurianum, a species of Moss.

Archidium Lescurii, a species of Moss.

Atrichum Lescurii, a species of Moss.

Fontinalis Lescurii, a species of Moss.

Hypnum Lescurii, a species of Moss.

Orthotrichum Lescurii, a species of Moss.

Sphagnum Lescurii, a species of Moss.

Thelia Lescurii, a species of Moss.

Alyssum Lescurii, a species of the Mustard Family.

Lepidophoroxs Lesquereuxii, a fossil plant.

Lepidodendrum Lesquereuxii, synonym of L. clypeatum, a fossil plant.

Sphenopteris Lesquereuxii, a fossil plant.

Sphaerites Lesquereuxii, a fossil plant.

Persoonia Lesquereuxii, a fossil plant.

Buthotrephis Lesquereuxii, a fossil plant.

Pecopteris Lesquereuxii, a fossil plant.

Cardiocarpon Lescurianum, a fossil plant.

Triphylopteris Lescuri, a fossil plant.

Odontopteris Lescurii, a fossil plant.
Sigillaria Lescurii, a fossil plant.
Sphenopteris Lescuriana, a fossil plant.
Taeniopteris Lescuriana, a fossil plant.
Triphyllopteris Lescuriana, a fossil plant.
Rhabdocarpus Lescurianus, a fossil plant.
Thinfeldia Lesquereuxiana, a fossil plant.
Laurinoxylon Lesquereuxiana, a fossil plant.

JOHN STRONG NEWBERRY.—Born at Windsor, Conn., Dec. 22, 1822, died in New York, Dec. 7, 1892. He graduated from Western Reserve College 1846 and from Cleveland Medical College 1848. He studied abroad for two years and then practiced medicine in Cleveland until 1855 when he was appointed acting Surgeon and Geologist to an exploring expedition through the country between San Francisco and the Columbia River. He went on several of these expeditions studying the botanical, zoological, and geological features of the country. In 1869 he was appointed Chief Geologist to the Geological Survey of Ohio.

In 1859 Newberry published the first State Catalogue of Ohio Plants. About 1865 he was made professor of Geology and palaeontology at Columbia College School of Mines which position he held until his death. His chief botanical work was in palaeo-botany though quite a number of living plants as well as many fossil species commemorate his name.

Newberrya, a genus of the Indian Pipe Family.
Gentiana Newberryi, a species of Gentian.
Abutilon Newberryi, a species of Malvaceae.
Leptosyne Newberryi, a species of Compositae.
Pentstemon Newberryi, synonym of *P. menziesii*, a species of Scrophulariaceae.
Potentilla Newberryi, synonym of *Ivesia gracillis*, a species of Rosaceae.
Ferula Newberryi, synonym of *Peucedanum Newberryi*, a species of Umbellifereae.
Astagolyous (Oxytropis) Newberryi, a species of the Leguminous Family.
Notholaena Newberryi, a species of Fern.

Physaria Newberryi, a species of Fern.
Coloptera Newberryi, a species of Fern.
Leucampyx Newberryi, a species of Fern.
Cardiocarpon Newberryi, a fossil plant.
Dadoxylon Newberryi, a fossil plant.
Odontopteris Newberryi, a fossil plant.
Cordaitea Newberryi, a fossil plant.
Pseudoplectopteris Newberryi, a fossil plant, synonym of
 Sphenopteris Newberryi.
Dictyophylon Newberryi, a fossil plant.
Cardiocarpus Newberryi, a fossil plant.
Archaeophylon Newberryanum, a fossil plant.
Viburnum Newberryanum, a fossil plant.
Celastrophyllum Newberryanum, a fossil plant.
Pecopteris Newberryana, a fossil plant.
Taeniopteris Newberryana, a fossil plant.
Platanus Newberryana, a fossil plant.
Laurus Newberryana, a fossil plant.
Myrica Newberryana, a fossil plant.
Salix Newberryana, a fossil plant.

H. C. BEARDSLEE.—Born in Connecticut. Died December, 1884, in Painesville, O. He came to Ohio and became a practicing physician at Painesville. He published the second State Catalogue of Ohio Plants in 1874. This was a pamphlet of nineteen pages; it was reprinted in the State Agricultural reports of 1877. After his death Dr. Beardslee's herbarium of about 4000 species, especially rich in Carices, Grasses, and Salices, was given to Oberlin College. The following species was named in his honor:

Chantransia violacea Beardslei, a species of fresh-water Alga.

ANDREW PRICE MORGAN.—Born at Centerville near Dayton, Oct. 27, 1836, now living in Hamilton County. He has done much toward the development of Mycology in this country. He has made known a large number of higher fungi of the south-eastern part of Ohio. He is the author of many new species. A number of plants belonging to the group of Fungi have been named in his honor, as follows:

Boletus Morgani, a species of Polyporeae.
Polyporus Morgani, a species of Polyporeae.
Lepiota Morgani, a species of Leucosporeae.
Russula Morgani, a species of Leucosporeae.
Cantharellus Morgani, a species of Leucosporeae.
Hypoxylon Morgani, a species of Pyrenomycetaceae.

There is also one species of Agaricineae named for Mrs. Morgan:

Hygrophorus Laurae, a species of Fungus.

WILLIAM ASHBROOK KELLERMAN.—Born at Ashville, Ohio May 1, 1850. He was educated at Cornell and the German Universities, receiving the degree of Ph.D. in 1881.

For a time he was teacher of natural science in the Wisconsin State Normal School, later was professor of Botany and Zoology at the Kansas Agricultural College, State Botanist of Kansas, and Botanist of the Kansas Experiment Station. He was the founder and editor (for four years) of the Journal of Mycology. He has held the position of Professor of Botany in the Ohio State University since 1890 and is the author of a number of text-books and articles for botanical journals. He has accumulated a very large and valuable private Herbarium of parasitic fungi, and a State Herbarium of the Flora of Ohio for the Ohio State University that already numbers many thousands of mounted sheets. The names given by botanists complimentary to his work are as follows:

Kellermannia, a genus of Sphaeropsideous fungi.
Aecidium Kellermannii, a species of Uredineae.
Plasmopora Kellermannii, a species of Phycomycetous Fungi.
Rosellinia Kellermannii, a species of Sphaeriaceous Fungi.
Rhabdospora Kellermannii, a species of Sphaeropsideous Fungi.
Diaporthe Kellermanniana, a species of Pyrenomycetous Fungi.
Physcomitrium Kellermani, a species of acrocarpous Moss.

MISS H. F. BIDDLECOME.—Of Columbus, formerly of Springfield, assiduously collected and studied the flora of Greene and Champaign

counties, Ohio. She discovered a species of Moss and one of Liverwort which have been named in her honor.

Bryum Biddlecomiae, a pleurocarpous Moss.

Trichocolea Biddlecomiae, a species of Hepaticae, or Liverwort.

F. D. KELSEY.—Born at New Washington, Ind., Feb. 15, 1849, but early moved to southern Ohio where he lived until 1856, when he moved to Columbus. He was graduated from Marietta College in 1870. The next year he went to Andover Theological Seminary where he graduated in 1874. He served Congregational Churches until 1885 when he moved to Helena, Montana. Here he made collections and distributed the local flora. The College of Montana conferred on him the degree of Sc. D.

In 1892 he was elected Professor of Botany at Oberlin College, which position he held until 1897, when he accepted the pastorate of the Central Congregational Church in Toledo, which he now holds. He is also lecturer on Botany at the Smead School for Girls at Toledo, Ohio. While professor at Oberlin he published several bulletins of local interest, and one monograph of *Uncinula*, with microscopic drawings of all American species. There are several species named in his honor, as follows:

Kelseya, a monotypic genus of Rosaceae.

Cucurbitaria Kelseyi, a species of Pyrenomycetous fungi.

Homostegia Kelseyi, a species of Pyrenomycetous fungi.

Cryptanthe Kelseyi, a species of Boraginaceae.

Phlox Kelseyi, a species of Polemoniaceae.

THE MAXIMUM HEIGHT OF PLANTS. II.

JOHN H. SCHAFFNER.

Last spring the writer published a few measurements of some common western plants in the Asa Gray Bulletin (Vol. 8: 19-20). A few other measurements are given below, all from Kansas except four which are from Ohio. It would be well for all collectors to keep records of the size of the plants which they meet, including measurements of the height and also dimensions of the leaves, flowers, fruits and underground parts. In this way the manual of the future may be brought to represent more accurately the living plant as it grows in nature rather than the dried, shrunk and dwarf specimens of the herbarium.

	Gray	Britton	Measured Feet
<i>Agropyrum repens</i> (L.) Beauv.		4	4¾
<i>Asparagus officinalis</i> L.		7	8¾
<i>Polygonum orientale</i> L. (Ohio)		8	9
<i>Allionia nyctaginea</i> Michx. (<i>Oxybaphus nyctagineus</i> Sw.)	3	3	5
<i>Silene antirrhina</i> L.	2½	2½	3
<i>Argemone alba</i> Lestib.			5
<i>Lepidium virginicum</i> L.			2½
<i>Baptisia leucantha</i> T. & G.		4	5½
<i>Amorpha canescens</i> Nutt.	3	3	4¾
<i>Kuhnistera candida</i> (Willd.) Kuntze (<i>Petalostemon candidus</i> Michx.)		2	3¼
<i>Acuan illinoensis</i> (Michx.) Kuntze (<i>Desmanthus brachylobus</i> Benth.)	4	3	8

Meriolix serrulata (Nutt.) Walp. (Oenothera serrulata Nutt.)	1¼	1½	2½
Gaura biennis L.	8	5	10½
Apocynum cannabinum L.	3		8
Marrubium vulgare L.		3	3½
Datura tatula L. (Ohio)		5	7
Verbascum thapsus L. (Ohio)		7	8
Cicuta maculata L.	6	6	7⅓
Symphoricarpos symphoricarpos (L.) MacM. (S. vulgaris Michx.)		5	8
Dipsacus fullonum L. (Ohio)		6	7½
Legouzia perfoliata (L.) Britt. (Specularia perfoliata A. D. C.)	1⅔	2	3
Ratibida columnaris (Sims) D. Don (Leptachys columnaris T. & G.)	2	2½	2⅝
Achillea millefolium L.		2	2½

CAMPUS BIRDS

A LIST OF BIRDS RECORDED BY THE WHEATON ORNITHOLOGICAL
CLUB, FOR THE O. S. U. FARM AND CAMPUS.

ROBERT F. GRIGGS.

The first list of Campus birds was published by the Wheaton Club in the "Agricultural Student" for March, 1898. Since that time a number of additional species have been reported and it has been thought advisable to republish the entire list, inserting the additions in their proper places. The list below contains 137 species. It is compiled from the records of the Wheaton Club, excepting those species for which Prof. J. R. Taylor is given credit. Only those species of which nests, eggs, or fledgelings have been observed are reported as breeding. The names are preceded by the A. O. U. check list number. Further additions will be published from time to time.

Ord. PYGOPODES.

Fam. PODICIPIDAE. GREBES.

6. *Podilymbus podiceps* (Linn.) Pied-billed Grebe, occasional.

Ord. ANSERES.

Fam. ANATIDAE. DUCKS, GEESE, etc.

137. *Anas americana* Gmel. American Widgeon, Bald-pate, occasional.
154. *Clangula hyemalis* (Linn.) Old squaw, occasional.
172. *Branta canadensis* (Linn.) Canada Goose, regular migrant.

Ord. HERODIONES.

Fam. ARDEIDAE. HERONS and BITTERNS.

- 190. *Botaurus lentiginosus* (Montag.) American Bittern.
- 191. *Ardetta exilis* (Gmel.) Least Bittern, accidental. (J. R. Taylor.)
- 194. *Ardea herodias* Linn. Great Blue Heron.
- 201. *Ardea virescens* Linn. Green Heron, common, breeds.

Ord. PALUDICOLAE.

Fam. RALLIDAE. RAILS, GALLINULES, etc.

- 221. *Fulica Americana* Gmel. Coot, occasional.

Ord. LIMICOLAE.

Fam. SCOLOPACIDAE. SNIPES, SANDPIPERS, etc.

- 228. *Philohela minor* (Gmel.) American Woodcock.
- 230. *Gallinago delicata* (Ord.) Wilson's Snipe.
- 256. *Totanus solitarius* (Wils.) Solitary Sandpiper, occasional.
- 263. *Actitis macularia* (Linn.) Spotted Sandpiper, common.

Fam. CHARADRIIDAE. PLOVERS.

- 273. *Ægialitis vocifera* (Linn.) Killdeer, common, breeds.

Ord. GALLINAE.

Fam. TETRAONIDAE. GROUSE, etc.

- 289. *Colinus virginianus* (Linn.) Quail, generally one or two flocks, breeds.

Ord. COLUMBAE.

Fam. COLUMBIDAE. PIGEONS and DOVES.

- 316. *Zenaidura macroura* (Linn.) Mourning dove, abundant, breeds.

Ord. RAPTORES.

Fam. CATHARTIDAE. AMERICAN VULTURES.

- 325. *Cathartes aura* (Linn.) Turkey Buzzard, occasional.

Fam. FALCONIDAE. HAWKS.

- 333. *Accipiter cooperi* (Bonap.) Cooper's hawk, occasional.
- 337. *Buteo borealis* (Gmel.) Red-tailed hawk, occasional.
- 343. *Buteo latissimus* (Wils.) Broad-winged hawk.
- 360. *Falco sparverius* Linn. American sparrow hawk, common, breeds.
- 364. *Pandion haliaetus carolinensis* (Gmel.) American Osprey,
accidental, reported by Mr. C. B. Morrey, during a spring flood.

Fam. BUBONIDAE. HORNED OWLS.

- 373. *Megascops asio* (Linn.) Screech Owl, common, breeds.
- 375. *Bubo virginianus* (Gmel.) Great Horned Owl, occasional.

Ord. COCCYGES.

Fam. CUCULIDAE. CUCKOOS, etc.

- 387. *Coccyzus americanus* (Linn.) Yellow-billed Cuckoo, common,
breeds.
- 388. *Coccyzus erythrophthalmus* (Wils.) Black-billed Cuckoo.

Fam. ALCEDINIDAE. KINGFISHERS.

- 390. *Ceryle alcyon* (Linn.) Belted King-fisher, common, breeds.

Ord. PICI.

Fam. PICIDAE. WOODPECKERS.

- 393. *Dryobates villosus* (Linn.) Hairy Woodpecker, common, resident.
- 394. *Dryobates pubescens* (Linn.) Downy Woodpecker, common,
resident, breeds.
- 402. *Sphyrapicus varius* (Linn.) Yellow-bellied Sapsucker, migrant.
- 406. *Melanerpes erythrocephalus* (Linn.) Red-headed Woodpecker,
common, breeds.
- 409. *Melanerpes carolinus* (Linn.) Red-bellied Woodpecker, resident.
- 412. *Colaptes auratus* (Linn.) Flicker, common resident, breeds.

Ord. MACROCHIRES.

Fam. CAPRIMULGIDAE. NIGHT-HAWKS, etc.

- 417. *Antrostomus vociferus* (Wils.) Whip-poor-will.
- 420. *Chordeiles virginianus* (Gmel.) Night-hawk.

Fam. MICROPODIDAE. SWIFTS.

- 423. *Chaetura pelagica* (Linn.) Chimney Swift, common, breeds.

Fam. TROCHILIDAE. HUMMINGBIRDS.

- 428. *Trochilus colubris* (Linn.) Ruby-throated Hummingbird.

Ord. PASSERES.

Fam. TYRANNIDAE. FLYCATCHERS.

- 444. *Tyrannus tyrannus* (Linn.) Kingbird, common, breeds.
- 452. *Myiarchus crinitus* (Linn) Crested Flycatcher.
- 456. *Sayornis phoebe* (Lath.) Phoebe, common.
- 461. *Contopus virens* (Linn.) Wood Pewee.
- 466a. *Empidonax traillii* (Aud.) Traill's Flycatcher, breeds.
- 467. *Empidonax minimus* Baird. Least Flycatcher. (J. R. Taylor.)

Fam. ALAUDIDAE. LARKS.

- 474. *Otocoris alpestris* (Linn.) Shore Lark, winter visitant.
- 474b. *Otocoris alpestris praticola* Hensh. Prairie Horned Lark.

Fam. CORVIDAE. CROWS, etc.

- 477. *Cyanocitta cristata* (Linn.) Blue Jay, common resident, breeds.
- 488. *Corvus americanus* Aud. Crow, seen at all seasons.

Fam. ICTERIDAE. BLACKBIRDS, etc.

- 494. *Dolichonyx oryzivorus* (Linn.) Bobolink, common.
- 495. *Molothrus ater* (Bodd.) Cowbird, common, breeds.
- 498. *Agelaius phoeniceus* (Linn.) Red-winged Blackbird.

- 501. *Sturnella magna* (Linn.) Meadow Lark, common, breeds.
- 506. *Icterus spurius* (Linn.) Orchard Oriole.
- 507. *Icterus galbula* (Linn.) Baltimore Oriole, common, breeds.
- 509. *Scolecophagus carolinus* (Müll.) Rusty Blackbird.
- 511b. *Quiscalus quiscula aeneus* (Ridgw.) Bronzed Grackle, Crow Blackbird, common, breeds abundantly.

Fam. FRINGILLIDAE. FINCHES and SPARROWS.

- 517. *Carpodacus purpureus* (Gmel.) Purple Finch.
- . *Passer domesticus* (Linn.) English Sparrow, superabundant, breeds.
- 529. *Spinus tristis* (Linn.) American Goldfinch, common.
- 533. *Spinus pinus* (Wils.) Pine Siskin.
- 540. *Pooecetes gramineus* (Gmel.) Vesper Sparrow, common, breeds.
- 542a. *Ammodramus sandwichensis savanna* (Wils.) Savannah Sparrow.
- 546. *Ammodramus savannarum passerinus* (Wils.) Grasshopper Sparrow, almost certainly breeds though no nest has been found.
- 552. *Chondestes grammacus* (Say.) Lark Sparrow.
- 554. *Zonotrichia leucophrys* (Forst.) White-crowned Sparrow.
- 558. *Zonotrichia albicollis* (Gmel.) White-throated Sparrow.
- 559. *Spizella monticola* (Gmel.) Tree Sparrow, common, winter resident.
- 560. *Spizella socialis* (Wils.) Chipping Sparrow, common, breeds.
- 563. *Spizella pusilla* (Wils.) Field Sparrow.
- 567. *Junco hyemalis* (Linn.) Snow-bird, common, winter resident.
- 581. *Melospiza fasciata* (Gmel.) Song Sparrow, abundant, resident, breeds.
- 583. *Melospiza lincolni* (Aud.) Lincoln's Sparrow.
- 584. *Melospiza georgiana* (Lath.) Swamp Sparrow.
- 585. *Passerella iliaca* (Merr.) Fox Sparrow.
- 587. *Pipilio erythrophthalmus* (Linn.) Towhee, common resident.
- 593. *Cardinalis cardinalis* (Linn.) Cardinal, common resident, breeds.
- 598. *Passerina cyanea* (Linn.) Indigo Bunting, common, breeds.

604. *Spiza americana* (Gmel.) Dickcissel.

Fam. TANAGRIDAE. TANAGERS.

608. *Piranga erythromelas* Vieill. Scarlet Tanager.

610. *Piranga rubra* (Linn.) Summer Tanager, May 4, 1899. (J. R. Taylor.)

Fam. HIRUNDINIDAE. SWALLOWS.

611. *Progne subis* (Linn.) Purple Martin, common.

613. *Chelidon erythrogaster* (Bodd.) Barn Swallow, common, breeds.

617. *Stelgidopteryx serripennis* (Aud.) Rough-winged Swallow.

Fam. AMPELIDAE. WAXWINGS.

619. *Ampelis cedrorum* (Vieill.) Cedar-bird, Cherry-bird.

Fam. LANIIDAE. SHRIKES.

622. *Lanius ludovicianus* Linn. Loggerhead Shrike.

Fam. VIREONIDAE. VIREOS.

624. *Vireo olivaceus* (Linn.) Red-eyed Vireo, breeds.

627. *Vireo gilvus* (Vieill.) Warbling Vireo, common spring migrant.

628. *Vireo flavifrons* Vieill. Yellow-throated Vireo.

629. *Vireo solitarius* (Wils.) Blue-headed Vireo.

Fam. MNIOTILTIDAE. WOOD WARBLERS.

636. *Mniotilta varia* (Linn.) Black and White Creeper, common migrant.

639. *Helminthus vermivorus* (Gmel.) Worm-eating Warbler.

641. *Helminthophila pinus* (Linn.) Blue-winged Warbler.

642. *Helminthophila chrysoptera* (Linn.) Golden-winged Warbler. (J. R. Taylor.)

645. *Helminthophila ruficapilla* (Wils.) Nashville Warbler. (J. R. Taylor.)

647. *Helminthophila peregrina* (Wils.) Tennessee Warbler.

- 648. *Compsothlypis americana* (Linn.) Parula Warbler.
- 650. *Dendroica tigrina* (Gmel.) Cape May Warbler.
- 652. *Dendroica aestiva* (Gmel.) Summer Warbler, common, breeds.
- 654. *Dendroica caerulescens* (Gmel.) Black-throated, Blue Warbler.
- 655. *Dendroica coronata* (Linn.) Yellow-rumped Warbler, Myrtle Warbler.
- 657. *Dendroica maculosa* (Gmel.) Magnolia Warbler.
- 659. *Dendroica pensylvanica* (Linn.) Chestnut-sided Warbler.
- 660. *Dendroica castanea* (Wils.) Bay-breasted Warbler.
- 661. *Dendroica striata* (Forst.) Black-poll Warbler, common fall migrant.
- 662. *Dendroica blackburniae* (Gmel.) Blackburnian Warbler.
- 667. *Dendroica virens* (Gmel.) Black-throated Green Warbler.
- 671. *Dendroica vigorsii* (Aud.) Pine Warbler.
- 672. *Dendroica palmarum* (Gmel.) Palm Warbler.
- 674. *Seiurus aurocapillus* (Linn.) Ovenbird.
- 675. *Seiurus noveboracensis* (Gmel.) Water-thrush.
- 676. *Seiurus motacilla* (Vieill.) Louisiana Water-thrush. (J. R. Taylor.)
- 677. *Geothlypis formosa* (Wils.) Kentucky Warbler.
- 678. *Geothlypis agilis* (Wils) Connecticut Warbler.
- 681. *Geothlypis trichas* (Linn.) Maryland Yellow-throat.
- 683. *Icteria virens* (Linn.) Yellow-breasted Chat, breeds.
- 684. *Sylvania mitrata* (Gmel.) Hooded Warbler.
- 685. *Sylvania pusilla* (Wils.) Wilson's Warbler. (J. R. Taylor.)
- 687. *Setophaga ruticilla* (Linn.) American Redstart, common migrant.

Fam. TROGLODYTIDAE. WRENS, etc.

- 704. *Galeoscoptes carolinensis* (Linn.) Cat-bird, common, breeds abundantly.
- 705. *Harporhynchus rufus* (Linn.) Brown Thrasher, common, breeds.
- 718. *Thryothorus ludovicianus* (Lath.) Carolina Wren, common resident, breeds.
- 721. *Troglodytes aëdon* Vieill. House Wren, common, breeds.

722. *Troglodytes hiemalis* Vieill. Winter Wren.

Fam. CERTHIIDAE. CREEPERS.

726. *Certhia familiaris americana* (Bonap.) Brown Creeper, common winter visitant.

Fam. PARIDAE. NUTHATCHES, etc.

727. *Sitta carolinensis* Lath. White-breasted Nuthatch, common resident.

728. *Sitta canadensis* Linn. Red-breasted Nuthatch, migrant.

731. *Parus bicolor* (Linn.) Tufted Titmouse, common resident.

735. *Parus atricapillus* (Linn.) Chickadee.

Fam. SYLVIIDAE. KINGLETS, etc.

748. *Regulus satrapa* Licht. Golden-crowned Kinglet, common.

749. *Regulus calendula* (Linn.) Ruby-crowned Kinglet.

751. *Polioptila caerulea* (Linn.) Blue-gray Gnatcatcher.

Fam. TURDIDAE. THRUSHES, etc.

755. *Turdus mustelinus* Gmel. Wood Thrush, common, breeds.

756. *Turdus fuscescens* Steph. Wilson's Thrush, Veery.

757. *Turdus aliciae* Baird. Gray-cheeked Thrush.

758a. *Turdus ustulatus swainsonii* (Cab.) Olive-backed Thrush.

759b. *Turdus aonalaschkae pallasii* (Cab.) Hermit Thrush.

761. *Merula migratoria* (Linn.) American Robin, abundant, breeds.

766. *Sialia sialis* (Linn.) Bluebird, common.

MINOR PLANT NOTES. No. 1.

W. A. KELLERMAN.

An introductory paragraph to this series may be short as will be the notes that follow. Suffice therefore to say that from time to time the field jottings and short notes of observations on the plants of our State flora will be presented. Not only those made myself but others which may be kindly communicated to me for this purpose by botanists, amateurs, students and other observers will find a place in the series. An especial invitation is extended to pupils in our High Schools and to teachers throughout the State to contribute suitable material which in many cases may be of great interest and importance. Sub-heads will be used thus making it possible to scan the notes easily.

PUCCINIA SMILACIS.—This Black Rust which has the various species of Green Briar for its host, was noticed in extraordinary quantity the past season in Scioto, Adams and Brown Counties. It was abundant only on *Smilax glauca*—called in that region the Saw Brier—but not uncommonly occurred on *S. hispida* and *S. rotundifolia*. It was also remarkable in this respect namely that the teleutospores (the black or winter spores) were excessively abundant: the uredospores (red or summer spores) could be found only when search was made. The observation here recorded pertains to the first half of the month of November. I have occasionally noticed uredospores in quantity on *Smilax* leaves in the various parts of Ohio but never before found the winter-spores common. It should perhaps be added parenthetically that *Dicæoma* having the priority over *Puccinia* is really the generic name that should be used, but the commoner designation may here be tolerated.

QUERCUS ACUMINATA.—The Chestnut or Yellow Oak (name in Gray's Manual *Quercus Muhlenbergii*) is described as "a tall tree with thin *flaky* bark" (Gray), or "a tree with gray *flaky* bark" (Britton); and in fact many of the White Oaks are said to have "*flaky*" bark. The *Quercus acuminata* is

very abundant in central Ohio and is of common occurrence in many parts of the State. It is rare that the “flakiness” of the bark or a tendency to separate in plates, is pronounced or conspicuous in our Chestnut Oak trees. The single good example I have found is here shown in Fig. 1 reproduced from a photograph of a specimen growing in Hayden’s ravine near Columbus. Both trees shown are the same species but the smaller one has merely furrowed, not flaky, bark, and it is a fair representative of the trunks of this Oak as they occur in our vicinity. The usual form of bark is perhaps more clearly shown in Fig. 2, which is from a photograph of a large tree standing near the Horticultural Hall of the Ohio State University. This specimen has leaves remarkably narrow for *Q. acuminata* but the trunk is similar in appearance to the trees which have broad leaves.

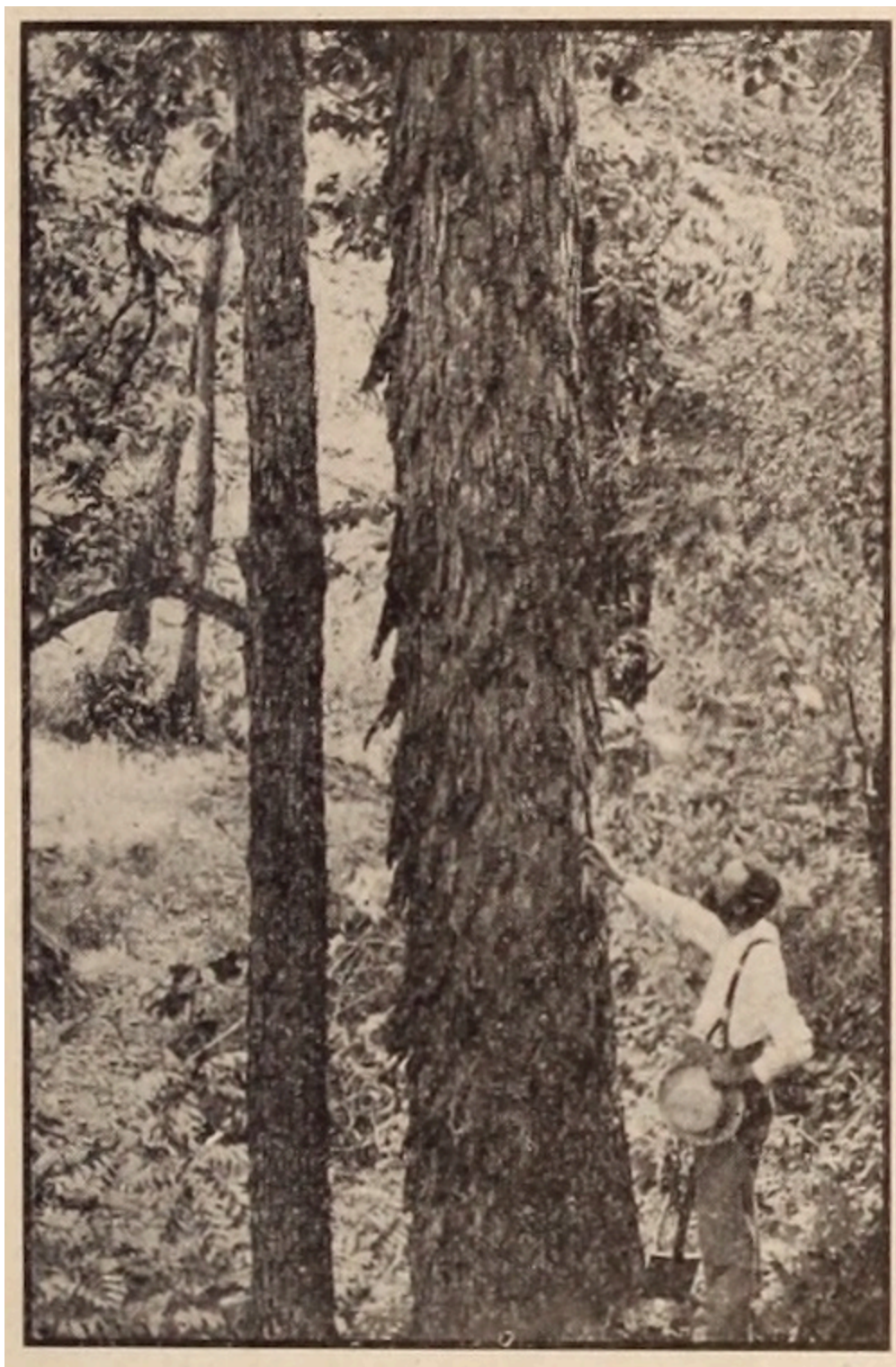


Fig. 1.

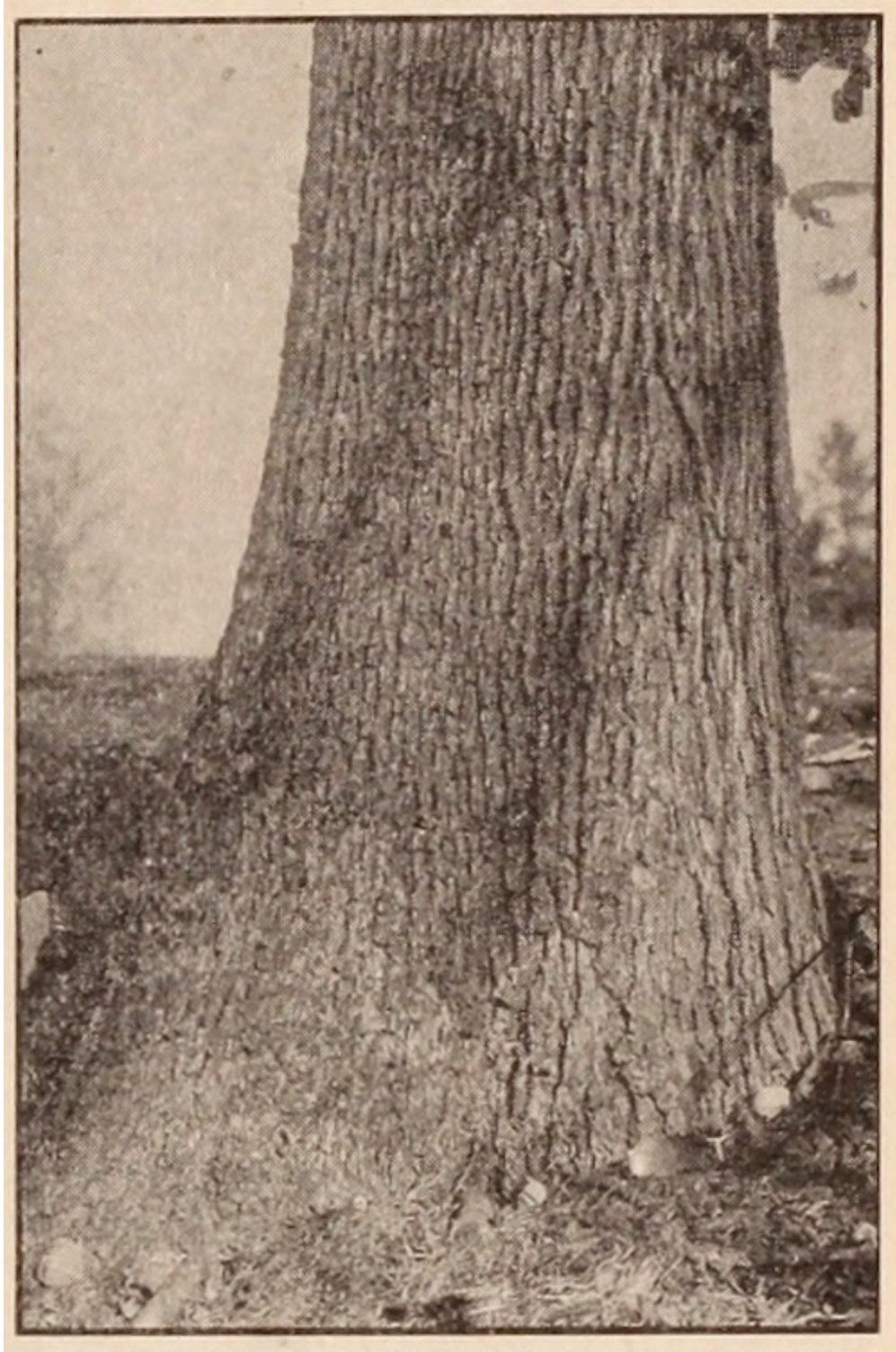


Fig. 2.

HELIANTHUS MAXIMILIANI.—Maximilian's Sunflower whose habitat and range are given in Britton and Brown's Illustrated Flora as "on dry prairies, Minnesota, and Manitoba to the Northwest Territory, Nebraska and Texas" may be said to be out of place in Ohio. Yet Mr. R. F. Griggs found it (a single plant) in the season of 1900 (cf. O. S. U. NATURALIST, 1:16) at Sandusky, Erie County. In a package of interesting plants sent to the State Herbarium recently by Mr. Otto Hacker, fine herbarium specimens of this conspicuous species were furnished. Mr. Hacker says that he observed it near Painesville (Lake County) six or seven years ago by railway tracks, where it is still flourishing.

LOTUS CORNICULATUS.—This European Lotus, with such common names as Bird's foot, Trefoil, Ground Honeysuckle, Bloom-fell, Crowtoes, Cat's-clover, and Sheepfoot, was collected at Painesville by Mr. Otto Hacker in nursery grounds. This is the second time this waif has been recorded for Ohio, the first case being reported by Mr. Ricksecker, of Oberlin.

NEWS AND NOTES.

Prof. Thomas A. Williams, editor-in-chief of the Asa Gray Bulletin and assistant chief of the Division of Agrostology, U. S. Department of Agriculture, died on the morning of Dec. 23, 1900.

At the 10th annual meeting of the Ohio State Academy of Science, held at Columbus, on December 26 and 27, thirty-one papers were read—12 pertaining to botany, 8 to zoology, 6 to geology, 3 to meteorology, and 2 to anthropology. The following are the officers for the coming year: President, August D. Selby, Wooster; 1st Vice Pres., Rev. H. Herzer, Marietta; 2nd Vice Pres, Mrs. W. A. Kellerman, Columbus; Secretary, E. L. Moseley, Sandusky; Treasurer, Herbert Osborn, Columbus; members of executive committee besides the president, secretary and treasurer, Thos. Bonser, Carey; Lynds Jones, Oberlin; trustees, F. M. Webster, Wooster; H. C. Beardslee, Cleveland; W. R. Lazenby, Columbus; publication committee, F. M. Webster, Wooster; John H. Schaffner, Columbus; L. H. McFadden, Westerville; Librarian, W. C. Mills, Columbus.

Since some criticism has been made, both at home and abroad, on the name of THE O. S. U. NATURALIST, it has been thought best to make a slight change by dropping the words, State University. Although a change in name is unfortunate it will not be very serious at the present time and hereafter the name will be THE OHIO NATURALIST, which is perhaps more descriptive of the scope of the journal and much better for purposes of citation.

J. H. S.

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This book was written in a period when many words had not become standardized in their spelling. Words may have multiple spelling variations or inconsistent hyphenation in the text. These have been left unchanged unless indicated below.

Obvious printing errors, such as backwards, upside down, reversed, or partially printed letters, were corrected. Final stops missing at the end of sentences and abbreviations were added.

The following spelling correction were made:

existance to [existence](#)

synonym to [synonym](#)

Onio to [Ohio](#)

miscroscopic to [microscopic](#)

Querqus to [Quercus](#)

Suppies to [Supplies](#).

*** END OF THE PROJECT GUTENBERG EBOOK THE OHIO
NATURALIST, VOL. 1, NO. 3, JANUARY, 1901 ***

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